INDEPENDENT PORTFOLIO / OCTOBER 2026

OpsGuide AI

Source-grounded incident guidance for cloud and infrastructure troubleshooting.

A working Python retrieval prototype that connects an incident question to ranked runbook evidence. Answers remain traceable to the selected source.

10	20	100%
SYNTHETIC RUNBOOKS	EVALUATION QUERIES	DEMO TOP-1 ACCURACY

FROM QUESTION TO EVIDENCE

01 **Ask a troubleshooting question**

Example: Kubernetes pod deployment image failure.

02 **Rank source material**

TF-IDF vectors + cosine similarity select relevant runbooks.

03 **Return the source or decline**

Extractive answer + source IDs; insufficient evidence triggers a refusal.

Demo results use 20 corpus-derived synthetic questions. This is not an independent production benchmark. No LLM generation is implemented.

Built to explain and verify

Python standard library / JSON API / HTML + CSS + JavaScript

The problem

Troubleshooting guidance scattered across procedures can be difficult to locate. A useful assistant must retrieve relevant evidence and expose its source rather than invent instructions.

What the prototype does

Searches 10 demo runbooks covering Linux, IAM, Kubernetes, APIs, networking, cooling, power, and retirement. Displays ranked source identifiers and similarity scores. Returns the strongest source verbatim when evidence passes a threshold.

Validation

Five automated unit tests pass. The actual HTTP API passes checks for source retrieval, unknown-topic refusal, empty-input validation, and cross-origin rejection. Top-1 retrieval and recall@3 are both 100% on 20 synthetic cases.

Run and demonstrate

Extract OpsGuide_AI.zip. Run **python app.py** and open **http://127.0.0.1:8001**. Show the Kubernetes example, switch to IAM, then ask about astronomy to demonstrate insufficient evidence. Run **python -m unittest -v**.

Scope and next steps

The current response is extractive, with lexical retrieval; no embeddings, LLM inference, AWS deployment, or agentic execution. Next steps: held-out evaluation, private-document ingestion, optional synthesis, and access controls. Browser rendering has not been verified in the build environment.

Career alignment
AI application engineering, retrieval grounding, API integration, cloud diagnostics, and response validation.